

4 (**short**) In a polytropic process, the pressure, p , and specific volume, v , of a perfect gas are related by $p v^n = \text{const.}$

(a) What types of process are represented by $n = \gamma$, 1, and 0. What is n for a constant volume process? [2]

(b) A gas in a cylinder expands against a fully-resisted piston in a polytropic process. Derive an expression for the work done by the gas in terms of its initial state (p_1, v_1) , its final state (p_2, v_2) , and n , where $n \neq 1$. [4]

(c) Sketch $p - v$ and $T - s$ diagrams of the process in (b) for

(i) $n = \gamma$,

(ii) $n = 1$,

from the same initial state (p_1, v_1) to the same final volume v_2 . [4]